

Stable matching

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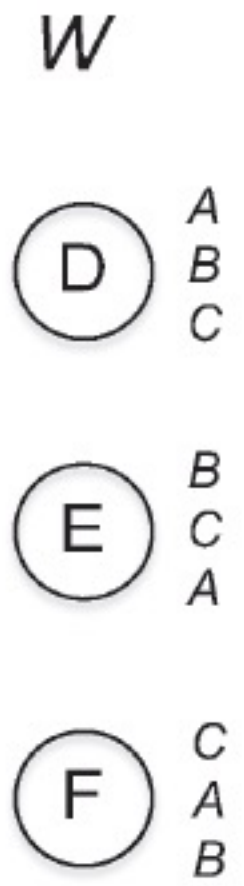
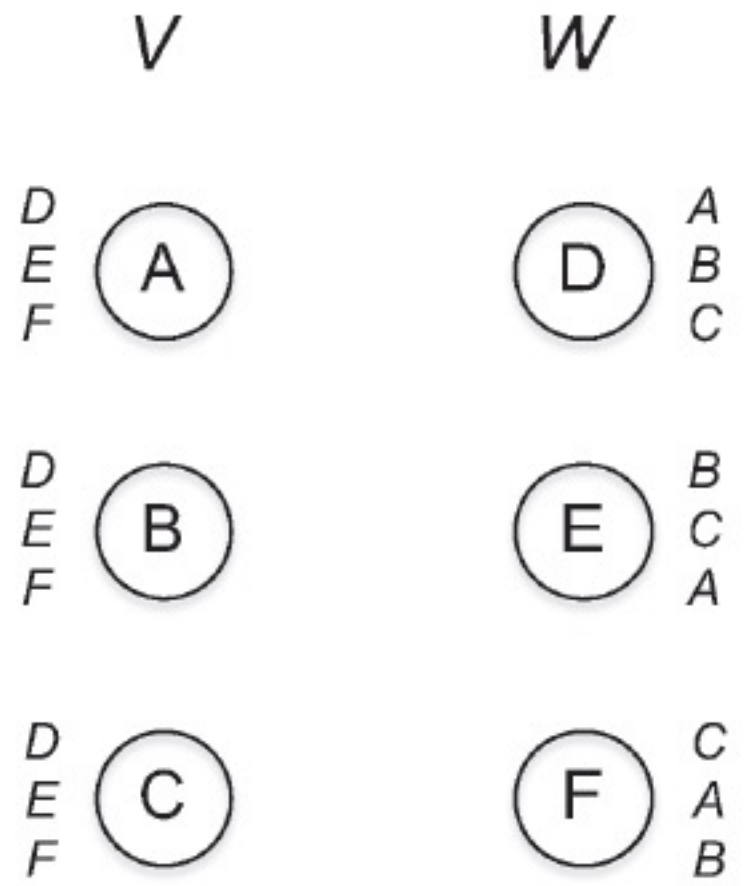
The stable marriage problem

- Given two sets of **equal size** (say males and females) we want to match each element of one set with one of the other set
- The problem arise because each element has preferences
- Common applications are
 - Matching students with schools
 - Kidney exchange
 - Other resources assignments

The game theoretic formulation

- Players: the members of both sets
- Actions: chose a member of the other set
- Preferences: decreasing order of the potential mates

Two visual representations of the problem



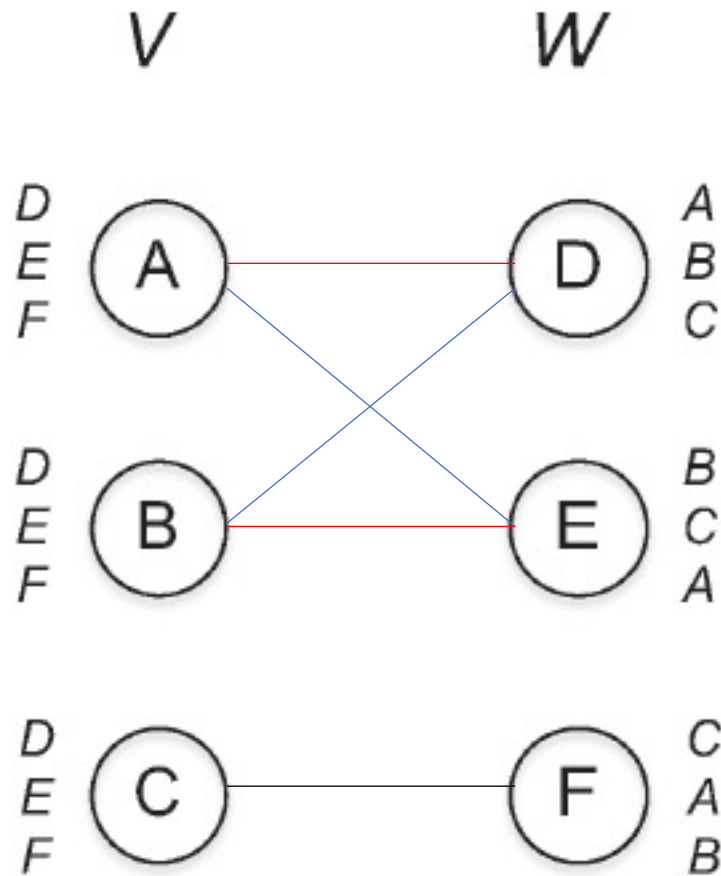
A	D,E,F
B	D,E,F
C	D,E,F

D	A,B,C
E	B,C,A
F	C,A,B

V

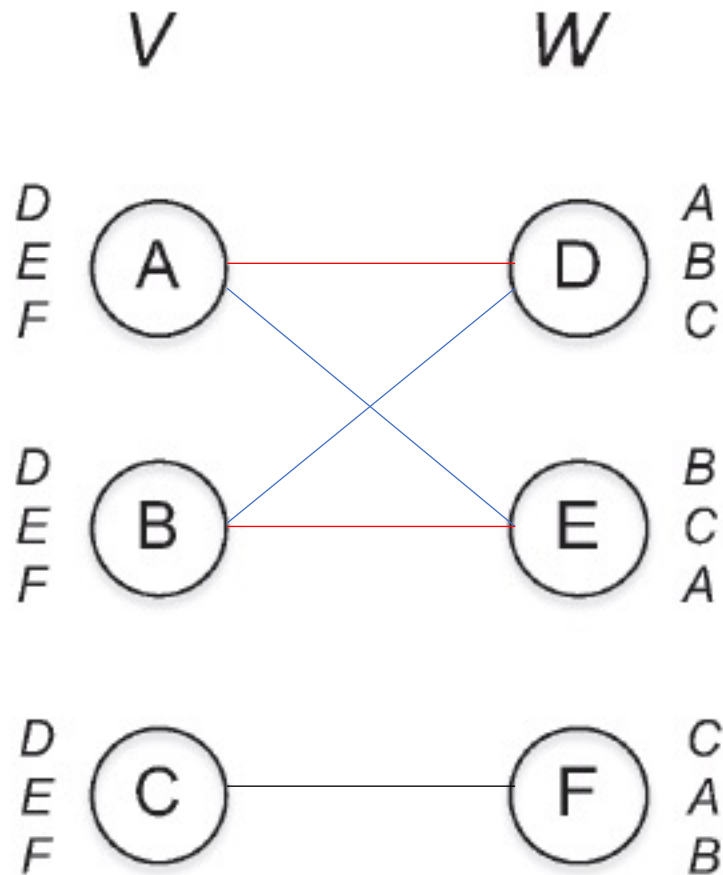
W

Observations on the Nash equilibrium



- A Nash equilibrium is a situation where there not exist two elements (one from V and one from W) that prefer to change their matching forming a new pair
- Consequently finding a Nash equilibrium is equivalent to finding a stable matching

Observations on the Nash equilibrium



A	D,E,F
B	D,E,F
C	D,E,F

V

D	A,B,C
E	B,C,A
F	C,A,B

W

Deferred Acceptance Algorithm (Gale-Shapley - 1962)

```
repeat until there are unmatched elements  $v_i \in V$   
  for  $i = 1$  to  $n$   
    repeat until  $v_i$  is not matched  
      pop  $w \in W$  from the preference list of  $v_i$   
      if  $w$  is unmatched  
        match ( $v_i, w$ )  
      else //  $w$  is matched with  $v_j$   
        if rank ( $w, v_i$ ) < rank ( $w, v_j$ )  
          match ( $v_i, w$ )  
          reject ( $v_j, w$ )  
        else  
          reject ( $v_i, w$ )
```

A practical example

	MALES
A	4 2 3 1 5
B	5 3 2 1 4
C	2 5 1 4 3
D	5 2 4 3 1
E	4 1 2 3 5

	FEMALES
1	D B E C A
2	B A D C E
3	A C E B D
4	D A C B E
5	B E A C D

A practical example

	MALES				
A	4	2	3	1	5
B	5	3	2	1	4
C	2	5	1	4	3
D	5	2	4	3	1
E	4	1	2	3	5

	FEMALES				
1	D	B	E	C	A
2	B	A	D	C	E
3	A	C	E	B	D
4	D	A	C	B	E
5	B	E	A	C	D

Matching: (A, 4), (B, 5), (C, 3), (D, 2), (E, 1)

Running time

- The algorithm takes $O(n^2)$ in the worst case where n is the number of elements in one of the two sets

Stability of the matching

- Consider a male M and a female F that are not matched in the final matching
- Case 1: M never attempted to match to F
 - Since M scroll his list from best to worst, M ended up matching with a better alternative
- Case 2: M attempted to match to F
 - he must have been rejected in favor of a male that F preferred (either at the time that F attempted to match to F , or subsequently).
 - Since the sequence of matched males only improves over the course of the algorithm, F ends up matched to a male she prefers

Other observations

- By construction the matching (i.e. the Nash equilibrium) always exists
 1. The two sets have the same cardinality,
 2. each male is matched only with a female at a time and vice versa
- Consequently for each male there must exist a free female
- The Nash equilibrium of Deferred Acceptance Algorithm is not necessarily the only equilibrium

Proposer optimality

- For a male M , let $h(M)$ denote the highest-ranked female (in M 's preference list) to which M is matched in any stable matching.
- The stable matching computed by the deferred acceptance algorithm matches every male M to $h(M)$

A practical example (using females as proposers)

	MALES
A	4 2 3 1 5
B	5 3 2 1 4
C	2 5 1 4 3
D	5 2 4 3 1
E	4 1 2 3 5

	FEMALES
1	D B E C A
2	B A D C E
3	A C E B D
4	D A C B E
5	B E A C D

A practical example (using females as proposers)

	MALES
A	4 <u>2</u> 3 1 5
B	<u>5</u> 3 <u>2</u> 1 4
C	2 5 1 4 <u>3</u>
D	5 2 <u>4</u> 3 1
E	4 <u>1</u> 2 3 5

	FEMALES
1	D B <u>E</u> C A
2	B <u>A</u> D C E
3	A <u>C</u> E B D
4	<u>D</u> A C B E
5	<u>B</u> E A C D

Matching: (A, 2), (B, 5), (C, 3), (D, 4), (E, 1)

Comparison using males vs females as proposers

	MALES				
A	4	2	3	1	5
B	5	3	2	1	4
C	2	5	1	4	3
D	5	2	4	3	1
E	4	1	2	3	5

	FEMALES				
1	D	B	E	C	A
2	B	A	D	C	E
3	A	C	E	B	D
4	D	A	C	B	E
5	B	E	A	C	D

Incentives

- The deferred acceptance algorithm is optimal only for proposers (males)
 - a) For every male M and reported preferences by the other males and females, M is never strictly better off reporting falsely than truthfully.
 - b) There exists preferences for a female F and reports for the other males and females such that F is strictly better off reporting falsely than truthfully.
- Ordering the females according to real preferences is a **dominant strategy for males**. The same is not true for females

A similar but different problem: the room mate problem

- In the room mate problem there is only one list of elements and we need to arrange them in stable pairs according to their preferences
 - **Example:** a group of students that need to be arranged in two bed rooms
- Game theoretic formulation:
 - **Players:** the list of elements to pair
 - **Actions:** chose a member of the list
 - **Preferences:** decreasing order of the potential mates
- Although the problem is conceptually similar and the solving algorithm (Irving's Algorithm) is similar, the Nash equilibrium for this problem may not exist

References

- Gale, David, and Lloyd S. Shapley. "College admissions and the stability of marriage." *The American Mathematical Monthly* 69.1 (1962): 9-15.
- Irving, Robert W. "An efficient algorithm for the “stable roommates” problem." *Journal of Algorithms* 6.4 (1985): 577-595.
- Video with the Irving’s algorithm
<https://www.youtube.com/watch?v=9Lo7TFAkohE>